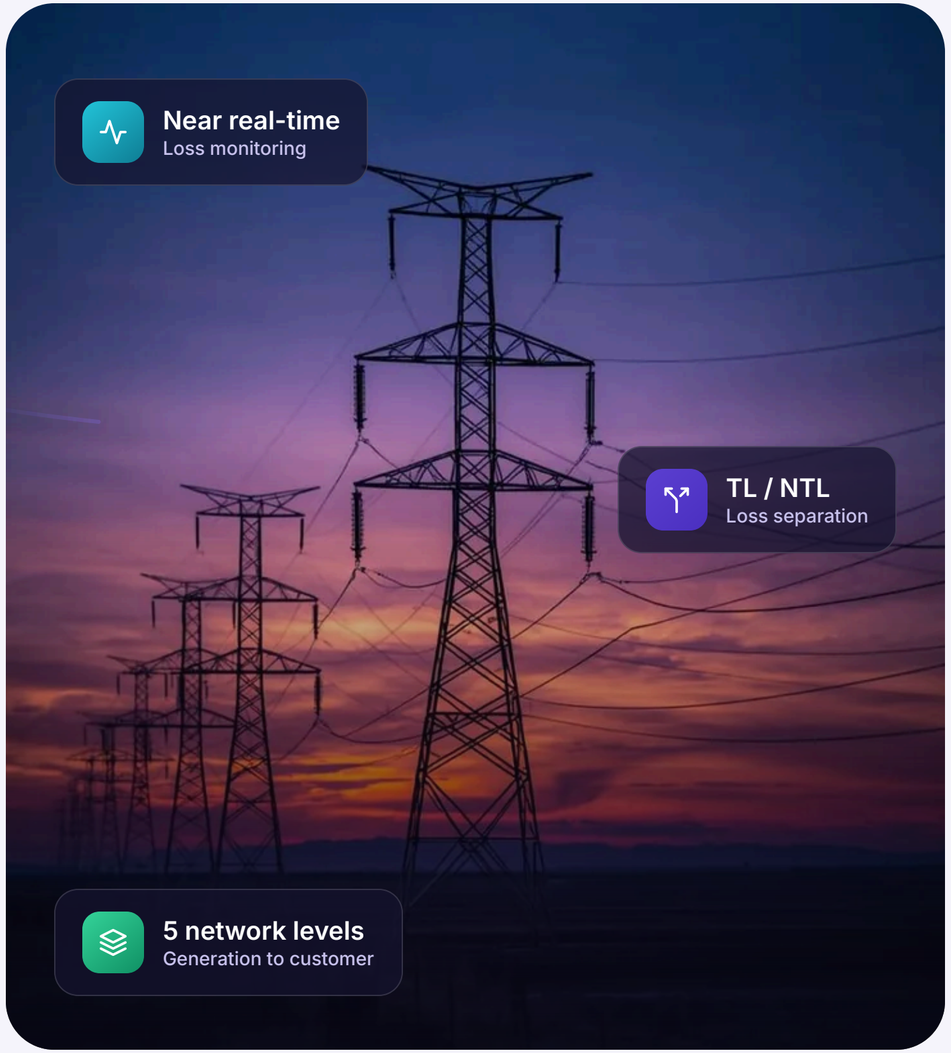
OONE POWER TECHNOLOGY · DEMO

Energy Loss Management System

ELMS for Electricité du Cambodge (EDC)

One integrated view to measure, separate and act on technical and non-technical energy loss across the distribution network.



Prepared by
OONE Power Technology Co., Ltd

AGENDA

Presentation overview

01 Current challenges

02 Project objectives

03 Solution overview

04 System architecture

05 Key benefits

06 Project timeline

07 Conclusion



01 · CURRENT CHALLENGES

No single view of energy loss

01

No dedicated loss monitoring system

Loss data is scattered across different systems with no single unified view.

It is hard to pinpoint exactly where losses occur across the distribution network.

02

Limited real-time visibility

Limited ability to detect losses at feeder, transformer and customer level.

Problem-solving is reactive, after the fact, rather than preventive.

03

Manual, disconnected analysis

Analysing and reconciling data takes a long time across separate systems, which delays detecting and resolving energy loss.

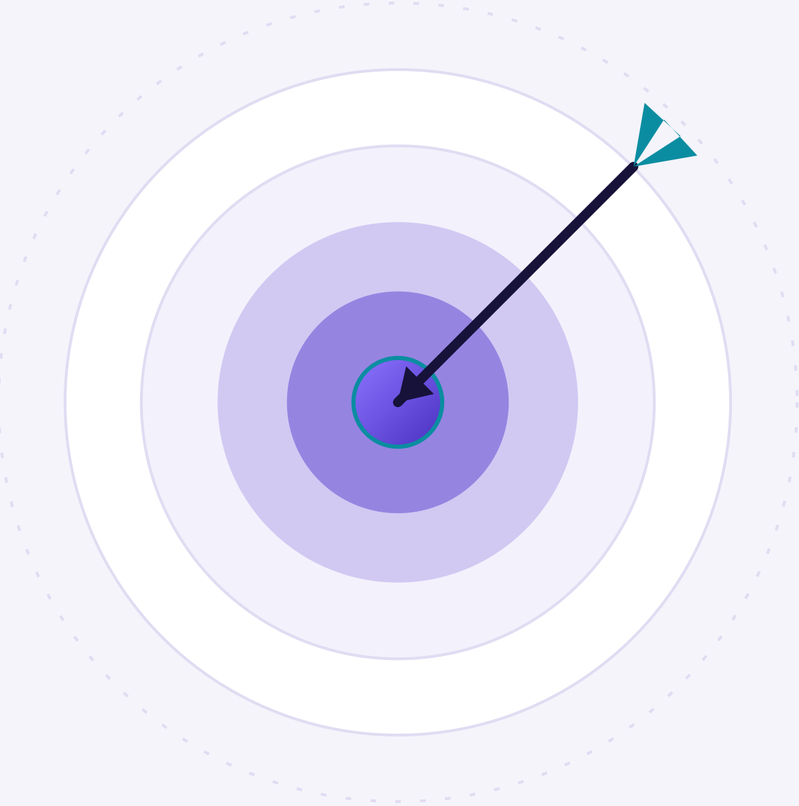
TODAY
Reactive response after incidents

WITH ELMS
Proactive detection and prevention

02 · PROJECT OBJECTIVES

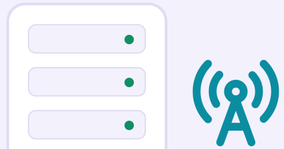
Five goals for energy loss management

1. More accurately calculate energy loss at each level of the network
2. Separate Technical Loss from Non-Technical Loss
3. Detect abnormal consumption and possible electricity theft early
4. Improve the accuracy of energy loss calculation
5. Support planning to improve the network and reduce losses



Existing systems ELMS builds on

Integration-first approach: ELMS connects to the existing OMMD system to protect current investment and minimise disruption to operations.



01 Multidrive HES

Collects Load Profile and meter data from smart meters, and records interval consumption data.

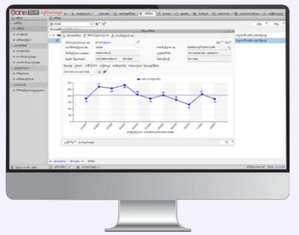
Head End System



02 OMMD Middleware

Connects billing with the Head End System. Collects and standardises Load Profile data and provides secure data exchange.

Central platform



03 OONE Billing

Manages customer billing, meter reading, invoices, accounts and usage history. Supplies billed energy for loss calculation.

Billed energy

03 · SOLUTION OVERVIEW

One module, one view of loss



Unified management through OMMD

ELMS is a module built inside OMMD, providing unified management and comprehensive loss monitoring across the whole network.

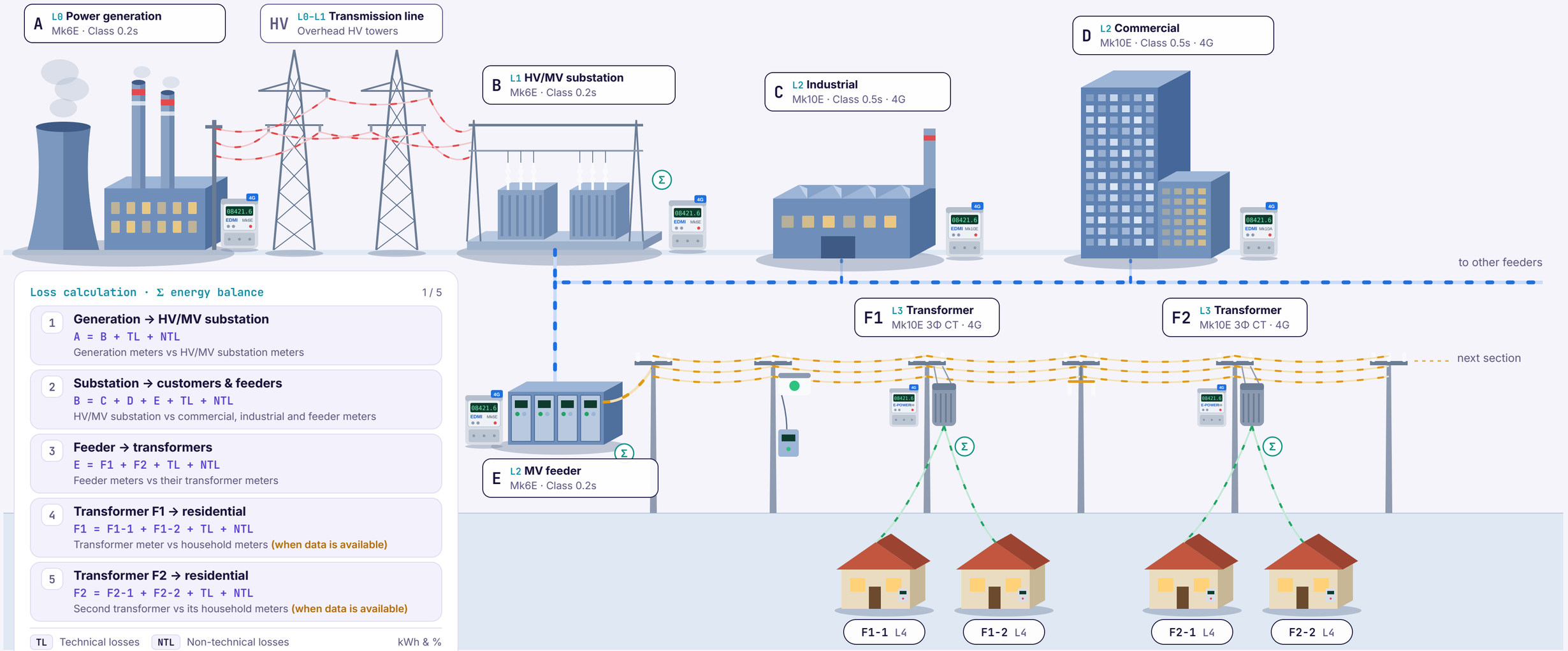
Automated analysis

Advanced automated analysis and near real-time monitoring detect anomalies early and continuously improve performance.

Single source of truth

Consolidated data keeps loss figures consistent across departments and supports decisions based on clear information.

Energy Loss Management Levels



Loss calculation · Σ energy balance 1 / 5

1. **Generation → HV/MV substation**
   `A = B + TL + NTL`
   Generation meters vs HV/MV substation meters
2. **Substation → customers & feeders**
   `B = C + D + E + TL + NTL`
   HV/MV substation vs commercial, industrial and feeder meters
3. **Feeder → transformers**
   `E = F1 + F2 + TL + NTL`
   Feeder meters vs their transformer meters
4. **Transformer F1 → residential**
   `F1 = F1-1 + F1-2 + TL + NTL`
   Transformer meter vs household meters **(when data is available)**
5. **Transformer F2 → residential**
   `F2 = F2-1 + F2-2 + TL + NTL`
   Second transformer vs its household meters **(when data is available)**

`TL` Technical losses `NTL` Non-technical losses kWh & %

The system will not only calculate energy losses but also use intelligent algorithms to help EDC identify and locate potential energy losses based on available data.

Zero consumption and abnormal inactivity detection

OBJECTIVE Identify meters reporting zero consumption for unusually long periods.

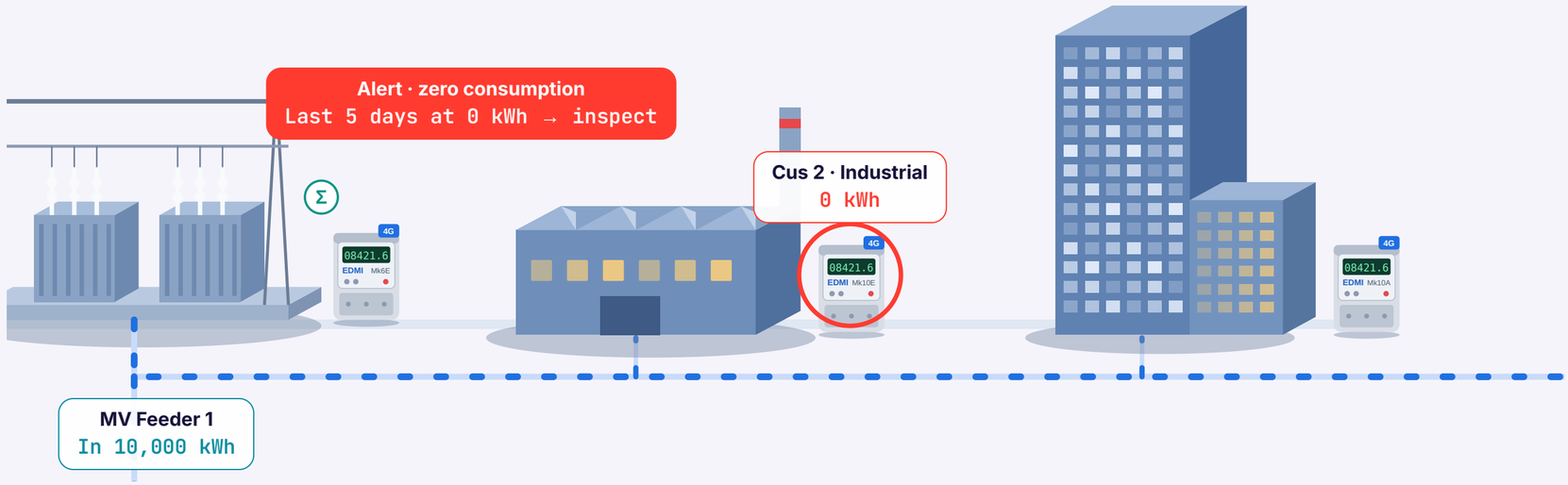
How the algorithm works

1 Feeder energy balance
MV Feeder 1 delivers 10,000 kWh, customers are billed 9,000 kWh → 1,000 kWh lost.

2 Read every meter
Cus 1 commercial 8,000 kWh · Cus 2 industrial 0 kWh · other customers 1,000 kWh.

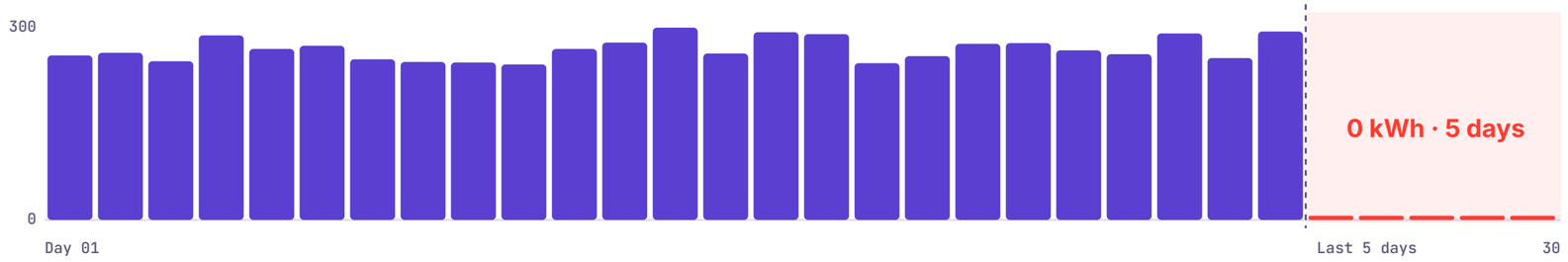
3 Zero for too long
Cus 2 has reported 0 kWh for the last 5 days in a row, after using ~270 kWh a day.

4 Flag for inspection
Possible meter bypass, stopped or faulty meter, or communication fault → field check.



Cus 2 · daily kWh

Example



Rule 0 kWh for ≥ N days (e.g. 3) · account active · feeder still supplying

Abnormal nighttime consumption detection

OBJECTIVE Detect consumers whose nighttime load behavior differs significantly from their normal usage.

How the algorithm works

1. **Learn the normal night**
   Each home's own 90-day profile: F2-1 normally uses ~0.3 kWh/h between 00:00 and 05:00.

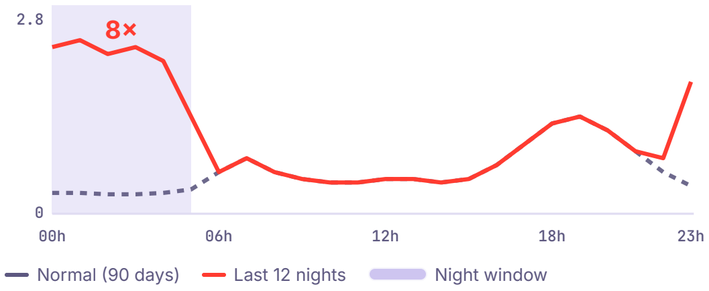
2. **Watch the night window**
   Every night, compare each home's 00:00–05:00 load with its own normal.

3. **Spot the difference**
   F2-1 now draws 2.4 kWh/h at night: 8× its normal, for 12 nights.

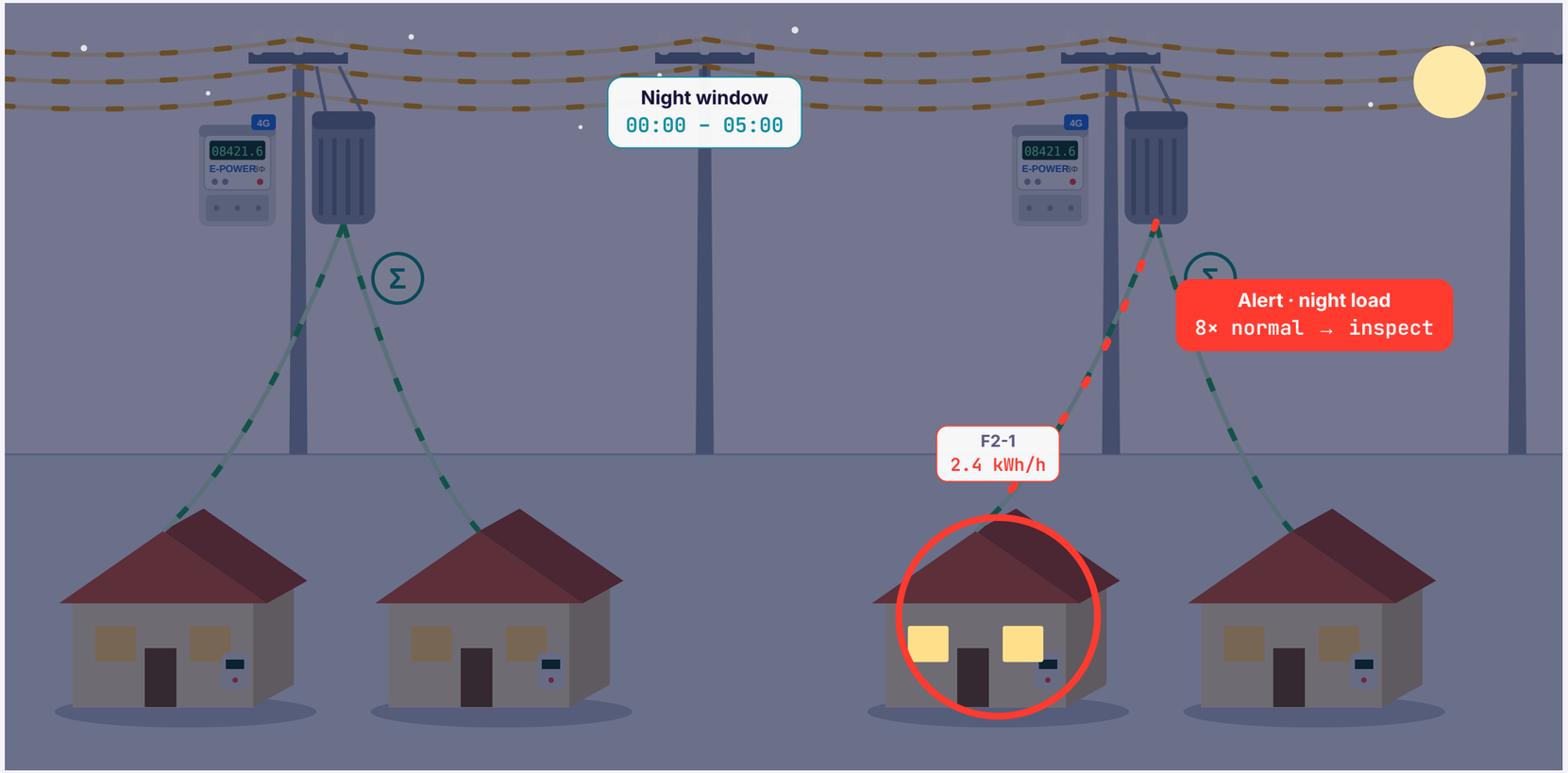
4. **Flag for inspection**
   Possible illegal connection, unauthorised use or meter tampering → field check.

F2-1 · 24-hour load profile

Example



Rule night load > X× own normal, several nights in a row

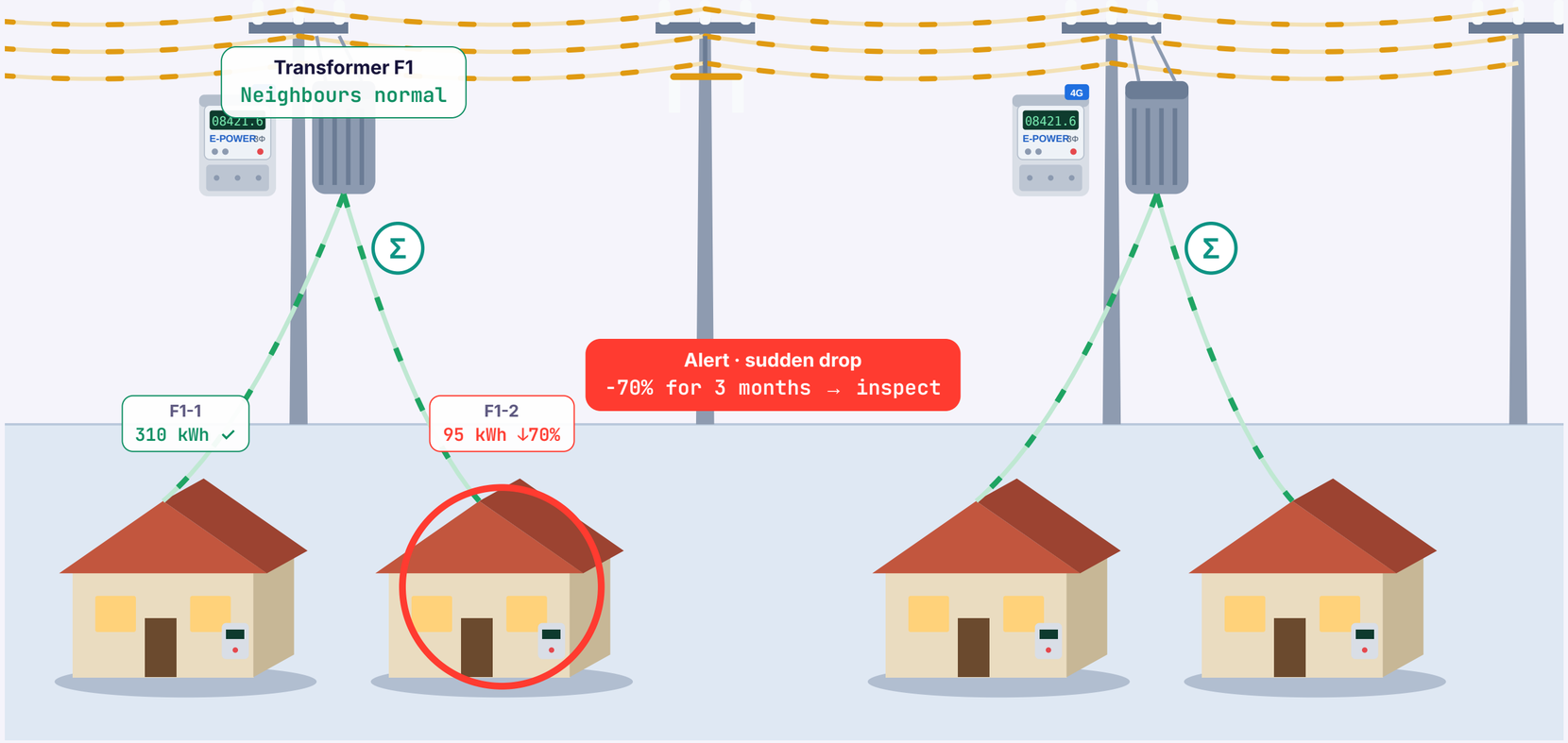


Individual consumer sudden consumption reduction

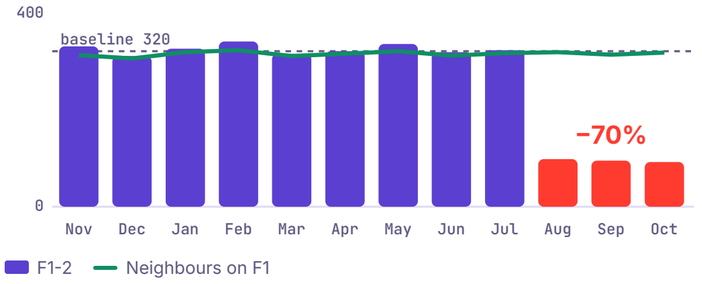
OBJECTIVE Find individual consumers whose electricity consumption suddenly decreases compared with their normal historical pattern.

How the algorithm works

1. **Build the history**
   12-month baseline per customer: F1-2 averages ~320 kWh a month.
2. **Compare recent months**
   Check each new month against that customer's own baseline.
3. **Sudden drop found**
   F1-2 falls to ~95 kWh a month (−70%) for 3 months in a row.
4. **Rule out & flag**
   Neighbours on transformer F1 stay normal, no season or vacancy reason → inspect for bypass or tampering.



F1-2 · monthly kWh — Example



Rule drop > Y% below own baseline for months · neighbours normal

03 · SOLUTION OVERVIEW · DASHBOARD

Dashboard and monitoring capability

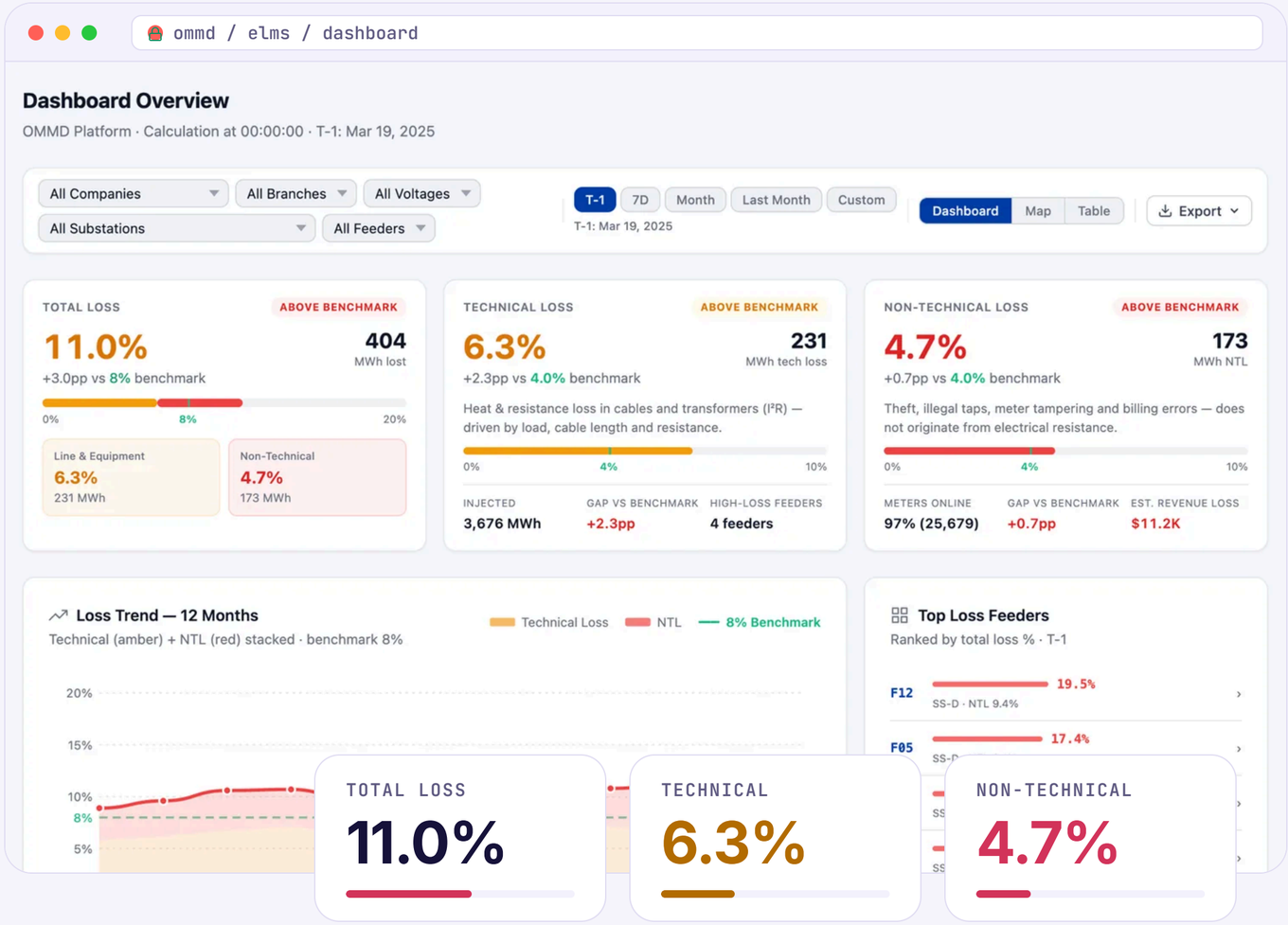
Sample figures from the ELMS demo

Near real-time data

LIVE

The dashboard instantly shows energy flow, losses and system performance at every network level.

- Total loss and Technical / Non-Technical split
- Energy loss analysis and trends
- Configurable threshold alerts
- Drill into high-loss feeders and substations
- Drill down by area, substation, feeder or transformer



Alerts and operational support

Threshold-based alerts

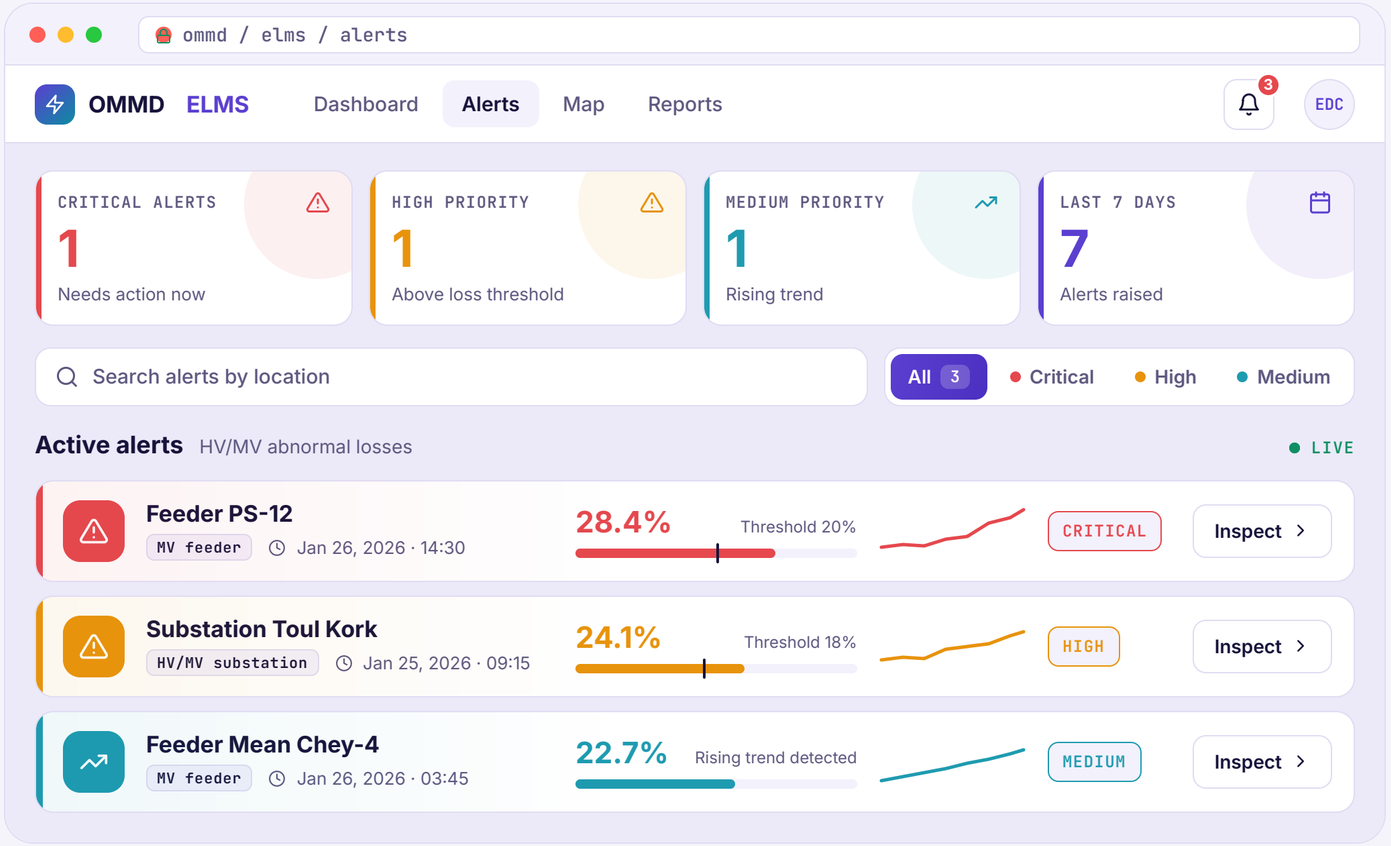
Alerts on loss percentage and abnormal consumption. The system sends notifications instantly.

Abnormal pattern detection

Identifies unusual consumption, signs of meter tampering and network irregularities.

Operational support

Key information and prioritised alerts help field teams go on site and resolve issues on time.



04 · SYSTEM ARCHITECTURE

From smart meter to final bill

ELMS is a module built inside OMMD, using OMMD as the central platform.

01 DATA SOURCE

Smart Meters



Interval data from every meter

4G Cellular · TCP/IP Ethernet

Load Profile →

02 HEAD END

Multidrive

MultiDrive

Head End System

Collects Load Profile

Load Profile →

03 PLATFORM

OMMD

Central middleware platform

Loss Management Module
Calculate on load profile data.

Final Reading →

04 BILLING

Oone Billing System

Oone BILLING SYSTEM

Final readings for invoicing

Customer invoices

1. Meters record interval data
2. Multidrive collects Load Profile
3. ELMS in OMMD calculates loss
4. Final reading goes to billing

05 · KEY BENEFITS

What ELMS delivers for EDC

Operational, financial and strategic benefits of the Energy Loss Management Module in OMMD.

01

Improved visibility of energy losses

- **Accurate, transparent** loss measurement
- **Energy balance** across generation, substations and feeders
- See **where and how** losses occur

02

Reduction of technical & non-technical losses

- **Early detection** of abnormal loss patterns
- **Targeted** corrective actions
- Direct **revenue protection** and recovery

03

Enhanced operational efficiency

- **Automates** manual, fragmented loss reporting
- Centralised **dashboards and alerts**
- **Faster** decisions and field investigations

04

Better use of existing investments

- Maximises value of **smart meters** and **MultiDrive HES**
- Builds on **OMMD middleware** and **OONE Billing**
- **No standalone platform** needed

05

Scalable foundation

- Ready for **advanced analytics**
- **Predictive loss detection**
- Supports **smart grid** and digital transformation

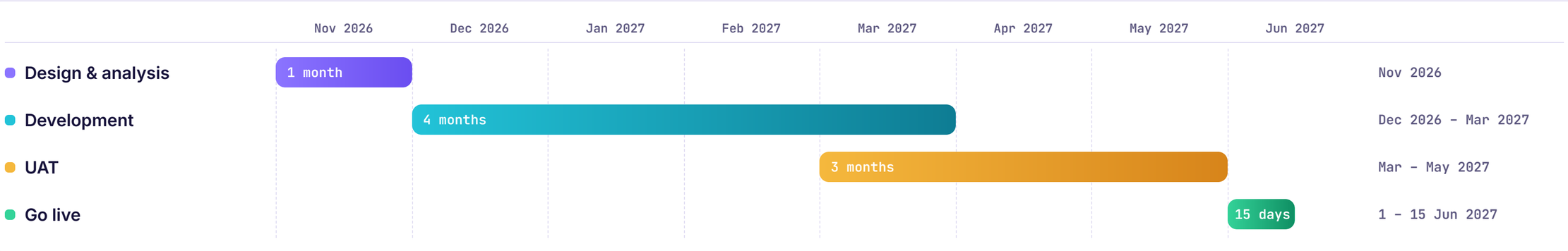
From generation to household: ELMS tracks energy flow down to each home **(when data is available for calculation)**, identifies loss points and supports effective loss-reduction strategies.

06 · PROJECT TIMELINE

Phased delivery over 8 months

Phased implementation with validation at every stage.

8
MONTHS TOTAL
Nov 2026 – Jun 2027



1 month

1 Design & analysis

- Case study and process analysis
- UI prototyping and approval
- Project kick-off

4 months

2 Development

- Build core features
- SIT and internal UAT
- First trial demo and bug fixing

3 months

3 UAT

- Second trial demo
- Pilot deployment and UAT training
- Final sign-off

15 days

4 Go live

- System setup
- Official start of operation

07 · CONCLUSION

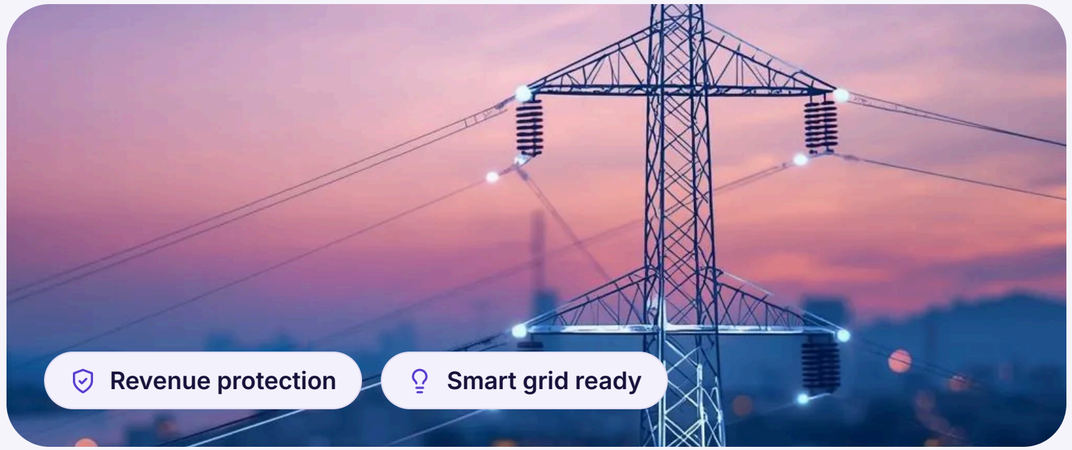
Conclusion

Summary of the proposed Energy Loss Management System for EDC.

A **comprehensive, data-driven** solution to monitor and reduce energy loss across EDC's network.

- Comprehensive **monitoring and analysis** of energy losses across the distribution network
- Built on **existing OMMD, OONE Billing and MultiDrive HES**: efficient to implement, maximising current infrastructure
- Tracks energy flow **from generation to household** **(when data is available for calculation)** and pinpoints loss points
- Supports **effective loss-reduction strategies** and protects EDC revenue

→ An important step toward a more **efficient, data-driven and modern** electricity distribution system.



ROLES & RESPONSIBILITIES

EDC

- Provide business requirements and approvals
- Facilitate system access and data availability
- Assign operational teams for loss investigation

OONE

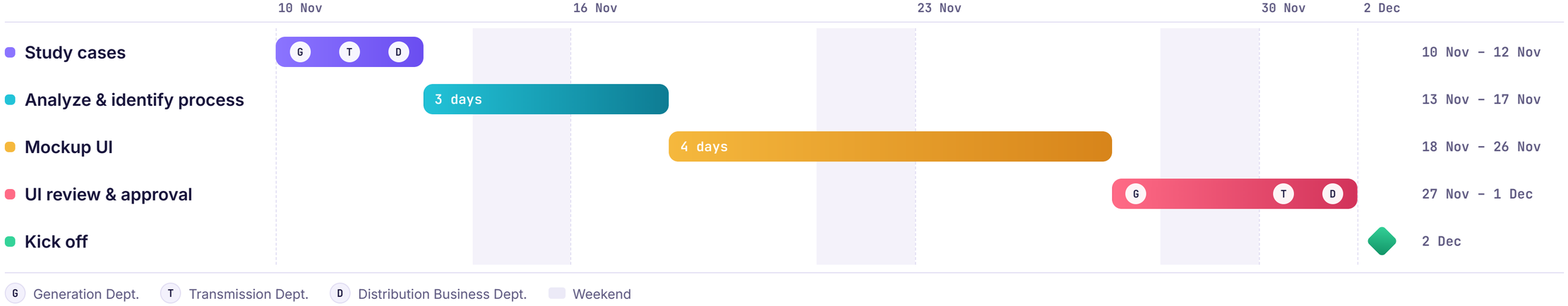
- Overall solution design and implementation
- System integration and configuration
- Training, documentation and post-go-live support

07 · CONCLUSION · NEXT STEPS

Next steps: design & analysis

First phase of the project, from study cases with each department to the official kick-off.

14
WORKING DAYS
10 Nov – 2 Dec 2026



3 days

1 Study cases

- Generation Dept. · 10 Nov
- Transmission Dept. · 11 Nov
- Distribution Business Dept. · 12 Nov

3 days

2 Analyze & identify process

- Analyse processes and identify operations · 13–17 Nov

4 days

3 Mockup UI

- Create the user interface mockup · 18–26 Nov

3 days

4 UI review & approval

- UI demo · Generation Dept. · 27 Nov
- UI demo · Transmission Dept. · 30 Nov
- UI demo · Distribution Business Dept. · 1 Dec

1 day

5 Kick off

- Project kick-off · 2 Dec 2026

POWER TECHNOLOGY

Thank you

Energy Loss Management System (ELMS) for EDC

Questions and discussion

Back to start

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